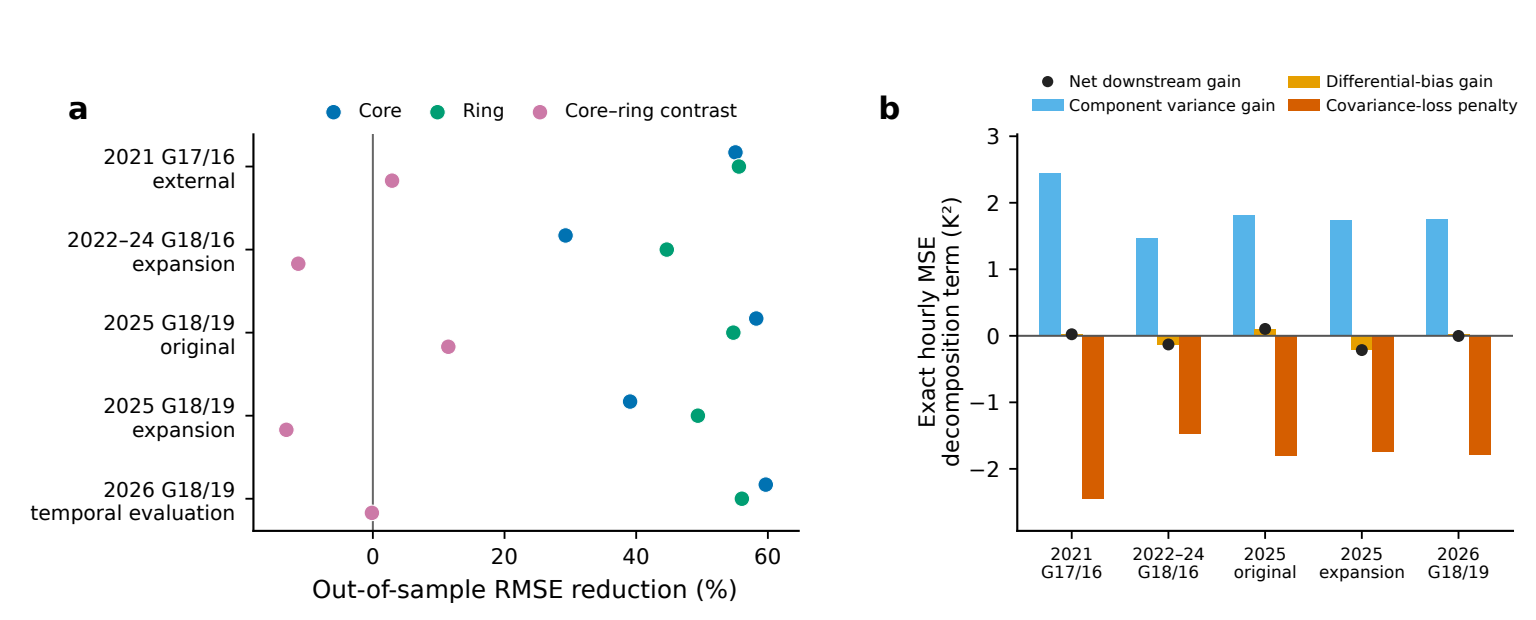
a
Core
Ring
Core–ring contrast
2021 G17/16
external
2022–24 G18/16
expansion
2025 G18/19
original
2025 G18/19
expansion
2026 G18/19
temporal evaluation
0
20
40
60
Out-of-sample RMSE reduction (%)
b
Net downstream gain
Component variance gain
Differential-bias gain
Covariance-loss penalty
Exact hourly MSE
decomposition term (K²)
3
2
1
0
−1
−2
2021
G17/16
2022–24
G18/16
2025
original
2025
expansion
2026
G18/19